

Shivam Shetty

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SUMMARY

AI/ML undergraduate with experience in AI, and full-stack systems. Built end-to-end ML pipelines and deployed production-grade applications using FastAPI, Docker, and Firebase. Hackathon winner with a focus on real-world AI systems including forest fire prediction and precision agriculture.

EDUCATION

2023 – 2027 B.E. in Artificial Intelligence and Machine Learning at **Shri Madhwa Vadiraja Institute of Technology, Udupi**

PUBLICATIONS

Enhancing STEAM Education Using Virtual Technologies: An Augmented Reality and Virtual Reality-Based Approach [View Paper](#)

Published in *IEEE Xplore*, 2026.

Swasthik Yesh, Manvitha Ravi Salian, Shivam Shetty, Gowri Bhat.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, Java, SQL

AI & Machine Learning: PyTorch, MONAI, TensorFlow, Keras, scikit-learn

Data: Pandas, NumPy, GDAL, Matplotlib, Seaborn, RasterIO, Xarray

Full-Stack & DevOps: FastAPI, Django, React, Next.js, Node.js, Express, Docker, Git

Databases & Cloud: PostgreSQL, MongoDB, Firebase, Supabase, Vercel

Specialized Tools: Unity, Streamlit, Postman, Roboflow

ACHIEVEMENTS

- **1st Place** – MONAITHON Hackathon, JNNCE, Shivamogga
- **1st Place** – Algorithm Roulette ML Competition, College Tech Fest
- **1st Place** – UI/UX Design Sprint, YUGMA TechFest, JNNCE
- **3rd Place** – HackYugma Hackathon, YUGMA TechFest, JNNCE
- **Best Project** – Lost and Found Portal (First-Year Mini Project)

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PROJECTS

Khandava – Satellite-Based Forest Fire Prediction

[GitHub](#)

- Architected a predictive system for Uttarakhand by integrating multi-source geospatial data from **NASA (FIRMS)**, **ISRO (Bhuvan, Bhoonidhi)**, and **ECMWF (ERA5)**.
- Optimized an **XGBoost** model to handle extreme class imbalance (450k+ samples), achieving a **75% Recall** for fire events, prioritizing early detection over global accuracy.
- Engineered preprocessing pipelines using **GDAL** and **python libraries** to preprocess high-resolution **ERA5** (meteorological), **DEM**, and **LULC** datasets.

Medical Image Segmentation (1st Place – MONAITHON)

[GitHub](#)

- Developed a **U-Net** segmentation model using the **MONAI** framework and **PyTorch** for automated retinal vessel extraction on the DRIVE dataset.
- Deployed the model as a production-ready application using **FastAPI** and **Streamlit** and **Docker**, focusing on low-latency inference.
- Secured **1st Place** at the JNNCE MONAITHON Hackathon for superior model precision and deployment workflow.

Coffee Ripeness Detection & Custom Dataset Engineering

[GitHub](#)

- Curated a custom dataset of **3,000+ high-resolution images** from plantation in Kalasa, Chikkamangaluru, capturing real-world variations in lighting, occlusion, and plant structure.
- Engineered a multi-class annotation strategy using **Roboflow** for coffee cherries and leaf structures to improve model contextual understanding.
- Trained and fine-tuned a **YOLO** object detection model to achieve high-precision identification of ripe vs. unripe cherries.
- Implemented data augmentation techniques to ensure model robustness against natural environmental noise and varying camera angles.

Department Voting Platform (120+ Users)

[GitHub](#)

- Designed and deployed a secure web-based voting system for departmental elections, handling **120+ voters** with real-time vote casting.
- Built the application using **Next.js** and **Firebase** (Authentication, Firestore) for scalable backend services.
- Implemented **Firebase Authentication** and role-based access control to ensure voter eligibility and prevent duplicate voting.
- Ensured data integrity and result transparency using **Firestore transactions** and controlled vote tallying mechanisms.

AR 3D Car Visualization (3rd Place – HackYugma)

[GitHub](#)

- Built a **Unity-based Augmented Reality** application for customizable real-time 3D vehicle visualization.
- Integrated interactive features for real-time mesh manipulation, material swapping, and lighting adjustments in AR.